Viruses

If you had DNA and RNA but could not make more copies of it

and

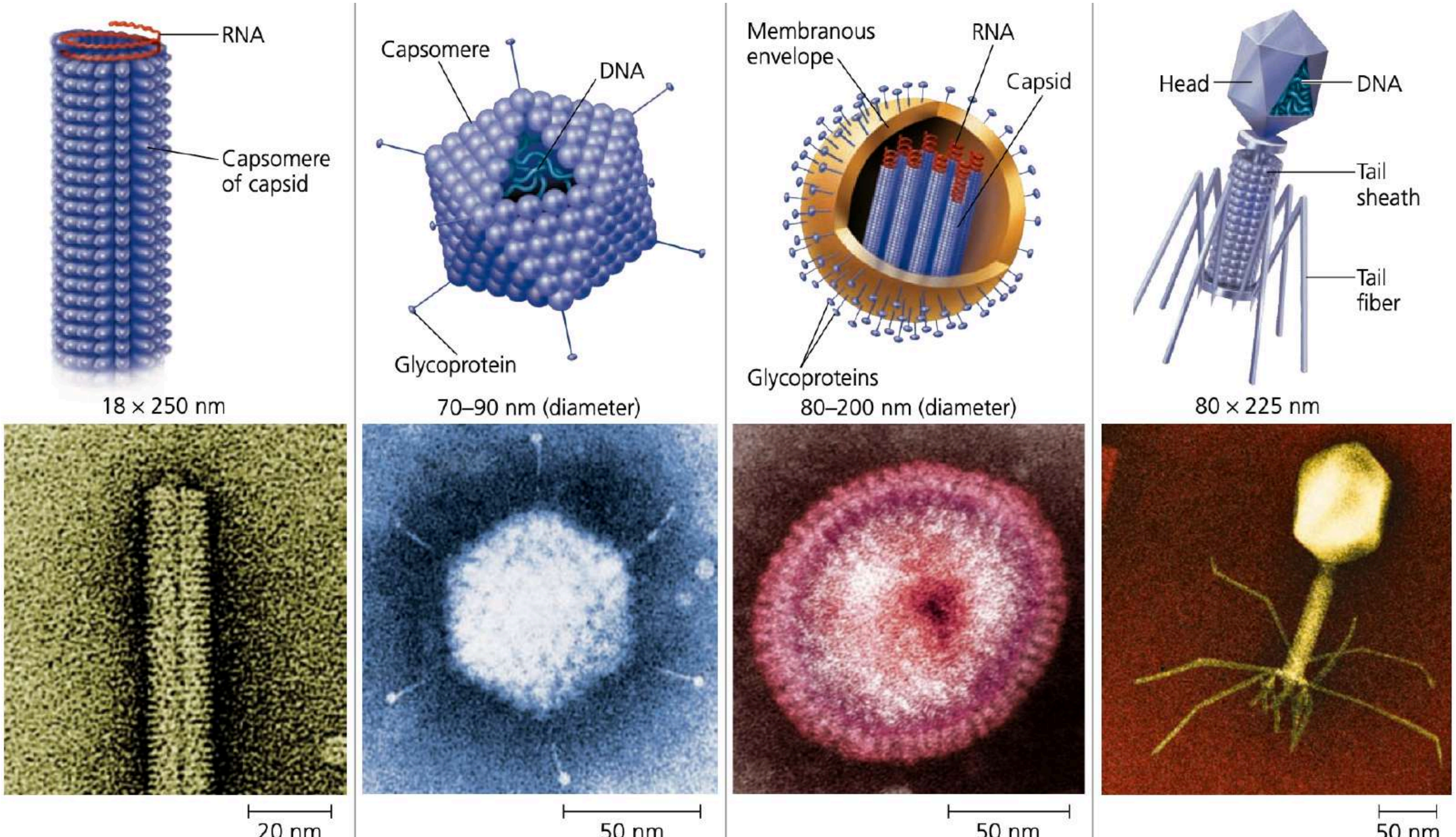
if your DNA and RNA codes for proteins

but

you cannot synthesize these encoded proteins

Are you "alive"?

Viruses



(a) Tobacco mosaic virus has a helical capsid with the overall shape of a rigid rod.

(b) Adenoviruses have an icosahedral capsid with a glycoprotein spike at each vertex (corner).

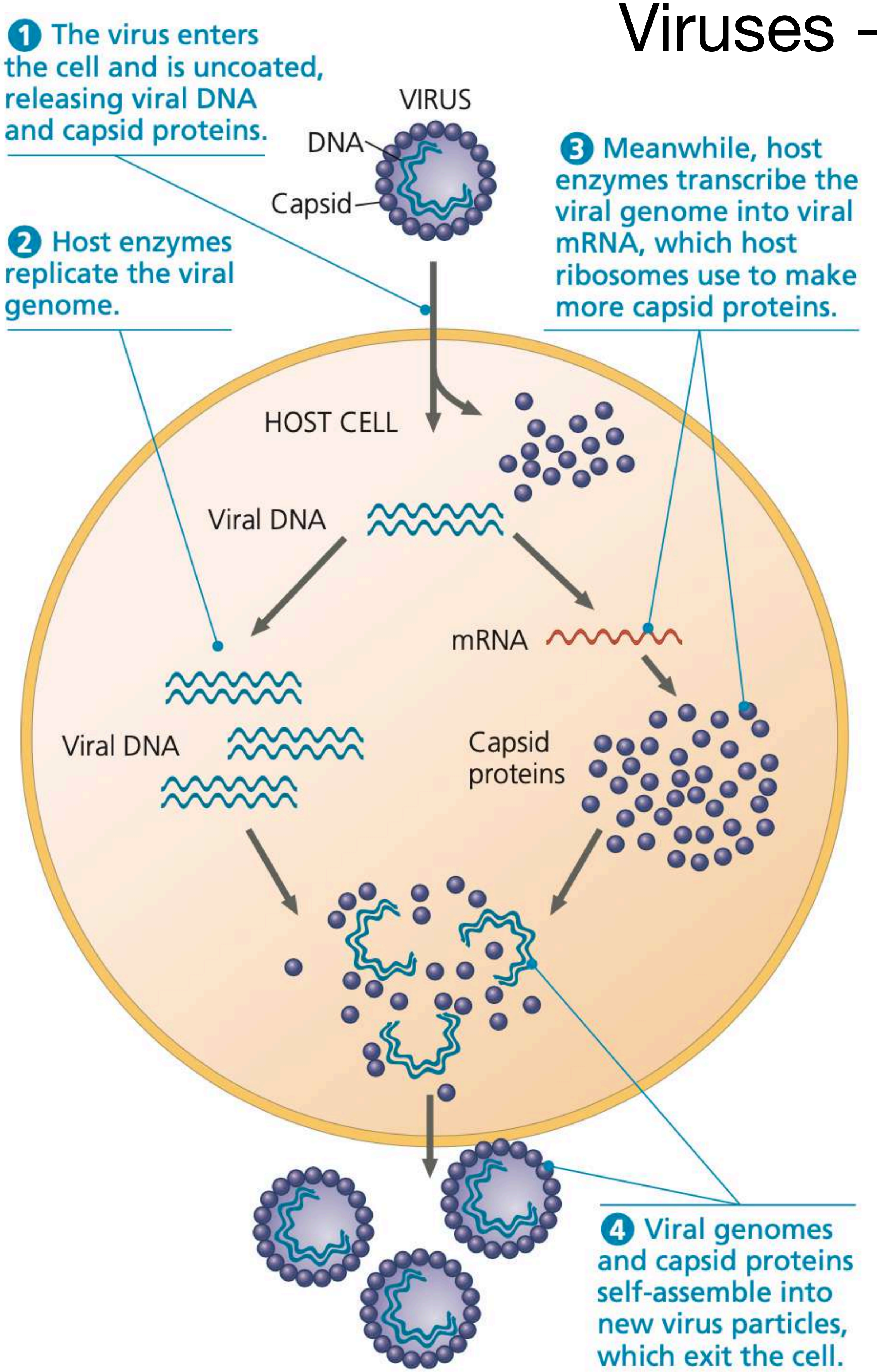
(c) Influenza viruses have eight different RNA molecules, each wrapped in a helical capsid, and an outer envelope studded with glycoprotein spikes.

(d) Bacteriophage T4, like other "T-even" phages, has a complex capsid consisting of an icosahedral head and a tail apparatus.

- Viruses contain genetic information as DNA or RNA, which is made into proteins by the machinery of the host cell
- If viruses require a host cell to come into existence, which came first, the virus or the host cell?
- Viruses infect ALL life forms: bacteria, plants, animals

Figure 19.3 of Campbell's Biology: a global approach

Viruses - no envelope

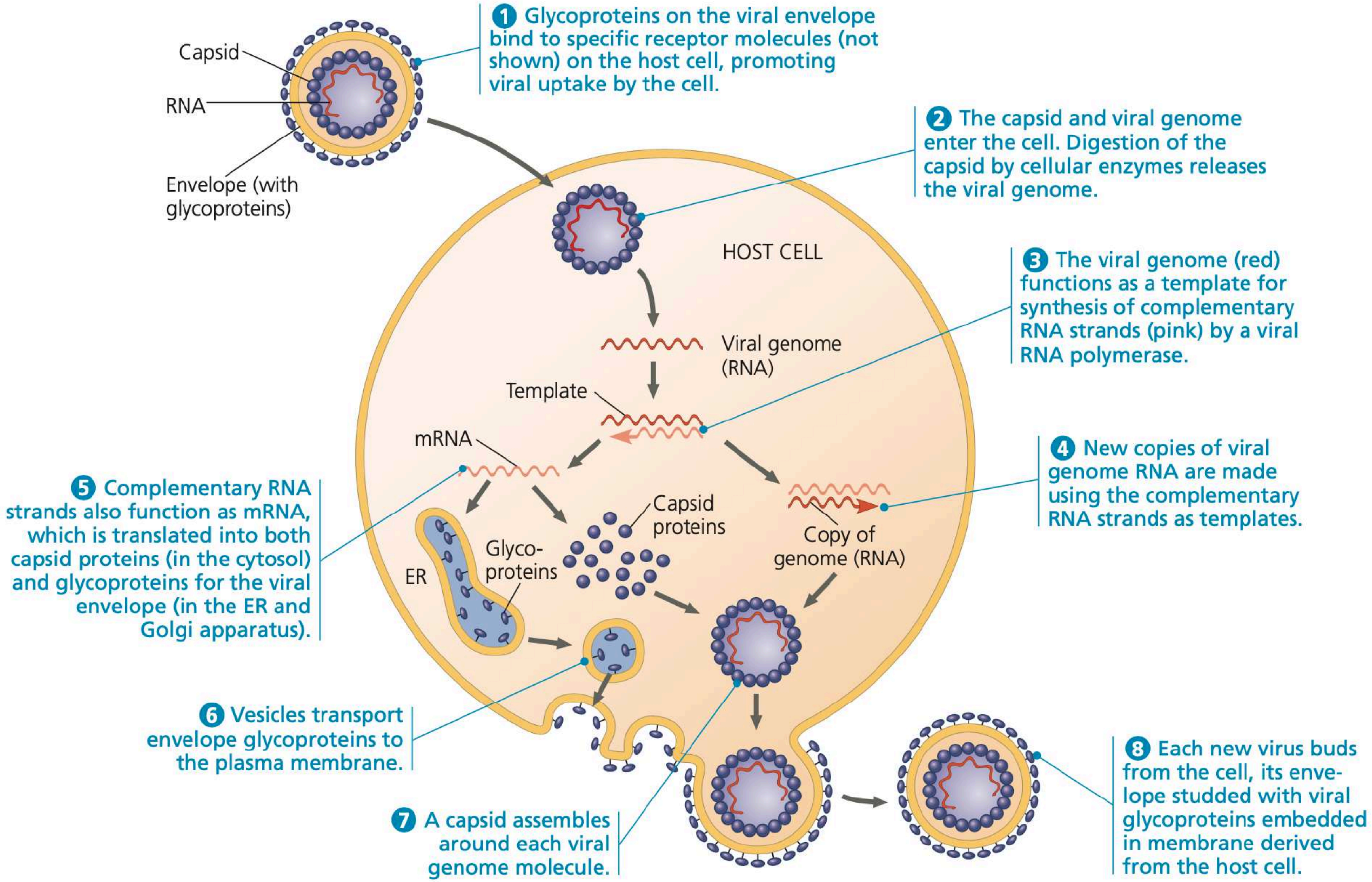


- To make more copies of itself, a virus particle known as a virion, has to enter a cell in which the machinery for genetic material replication and protein synthesis are functional.
- After it "hijacks" the host machinery and duplicates its genetic material, it uses the host machinery to package its genetic material.
- Then it ruptures the host cell and escapes into the host body, to find other cells into which it will enter and repeat the whole process.
- Viruses infect all life forms, but each type of virus can infect only a specific species of animal or a cell type in that animal.
- Excessive intrusion of humans into wild-life environments has caused an increase in the ability of viruses to cross species barriers.

Figure 19.4 of Campbell's Biology: a global approach

Sreelaja Nair

Viruses - with envelope



Genetic information in viruses can be DNA or RNA
DNA or RNA can be double or single stranded

Figure 19.7 of Campbell's Biology: a global approach

Viruses - vaccination

Prior infection with a milder variant provides immunity against subsequent exposure to more virulent strains

- Vacca = latin for cow
- Dr. Edward Jenner formally demonstrated that prior inoculation with cowpox provided protection from smallpox
- WHO declared smallpox as eradicated in 1980

Viruses - vaccination

- Louis Pasteur (1822–1895) initially showed attenuated pathogens provide protection against Cholera and Anthrax (bacteria) for farm animals

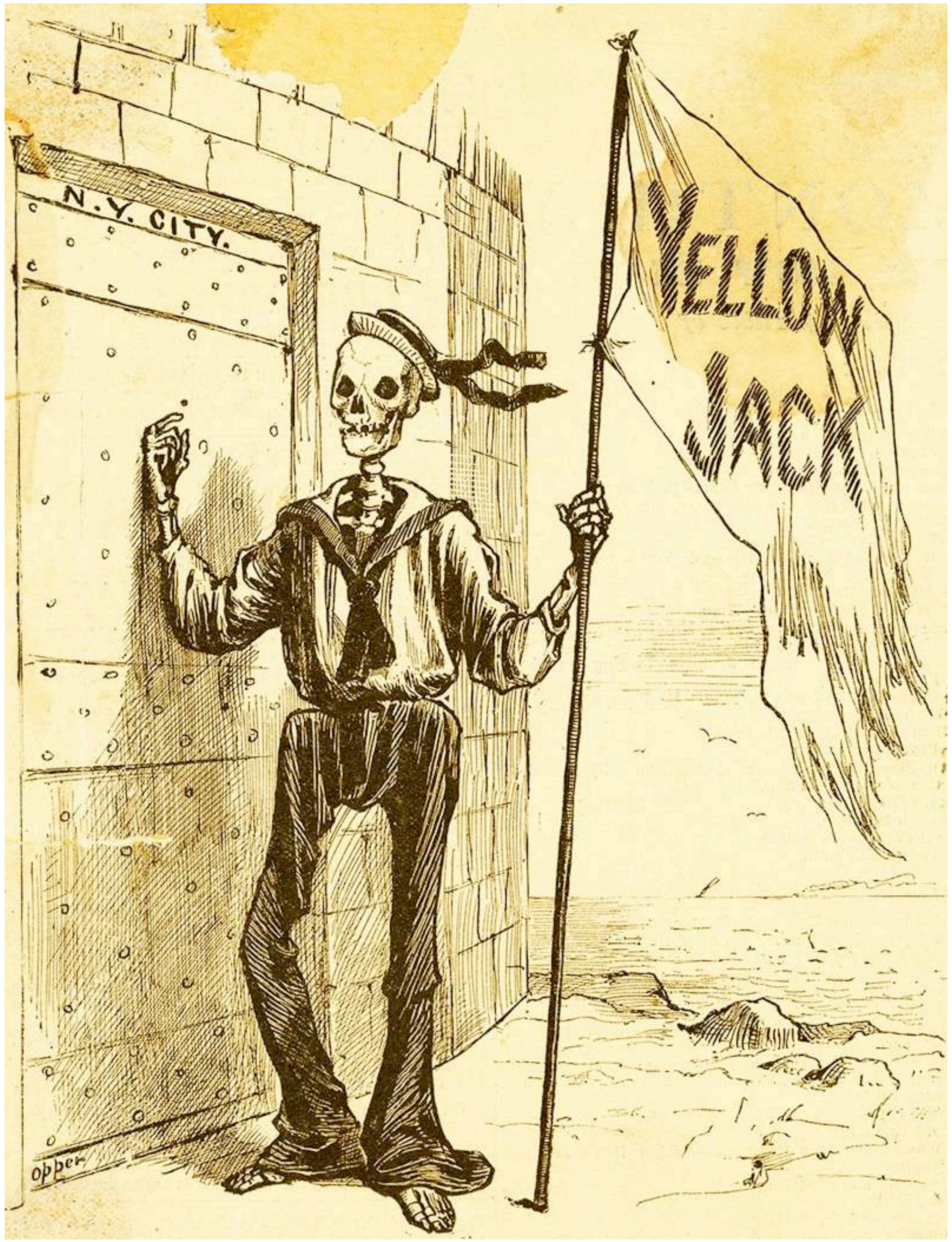
- In 1879, Pierre-Victor Galtier, a veterinary professor, serially transmitted rabies to rabbits from dogs
- ~1885, Pasteur used suspensions made from rabies-infected dried rabbit spinal cords to show that dogs could be protected from rabies infection

- Pasteur`s rabies strategy was post-exposure vaccine prophylaxis , still used today

- Pasteur found the strategy to vaccinate without ever isolating the causative agent, which would eventually be recognized in the twentieth century to be nonbacterial

Because rabies was 100% fatal, Pasteur went ahead and “vaccinated” humans bitten by rabid dogs without proper animal trials

Viruses - vaccination



- End of 19th century, yellow fever, caused by a flavivirus spread by infected female *Aedes aegypti* mosquitoes, had spread around the world due to slave trading and global markets
- Ships were required to fly the yellow quarantine flag - main symptom is jaundice
- South African virologist Max Theiler at the Harvard University School of Tropical Medicine, USA showed that the disease was viral
- Theiler showed that repeated passage in mouse brain cultures reduced the effect of the virus on most organs, but potentially increased its impact on the central nervous system, which could cause encephalitis
- Individuals could be vaccinated, but with risk of neurotoxicity

Viruses - vaccination

- In 1937, Max Theiler and Hugh Smith developed live attenuated yellow fever vaccine strain 17D: 176 serial passages in mouse embryonic tissue, then monkey serum, then minced whole chick embryo and then in chick embryo from which the brain and spinal cord had been removed.
- After these passages, 17D lost neurotropism, viscerotropism and mosquito competence, but still triggered an immune response in host

- Theiler showed that the Yellow Fever 17D vaccine prepared from infected whole chick embryos was safe and effective for human use without the addition of human serum - made it cost-effective and easier to produce in large quantities

- The Yellow Fever 17D vaccine was approved for human use in 1938 and provides lifelong protection with a single vaccine dose

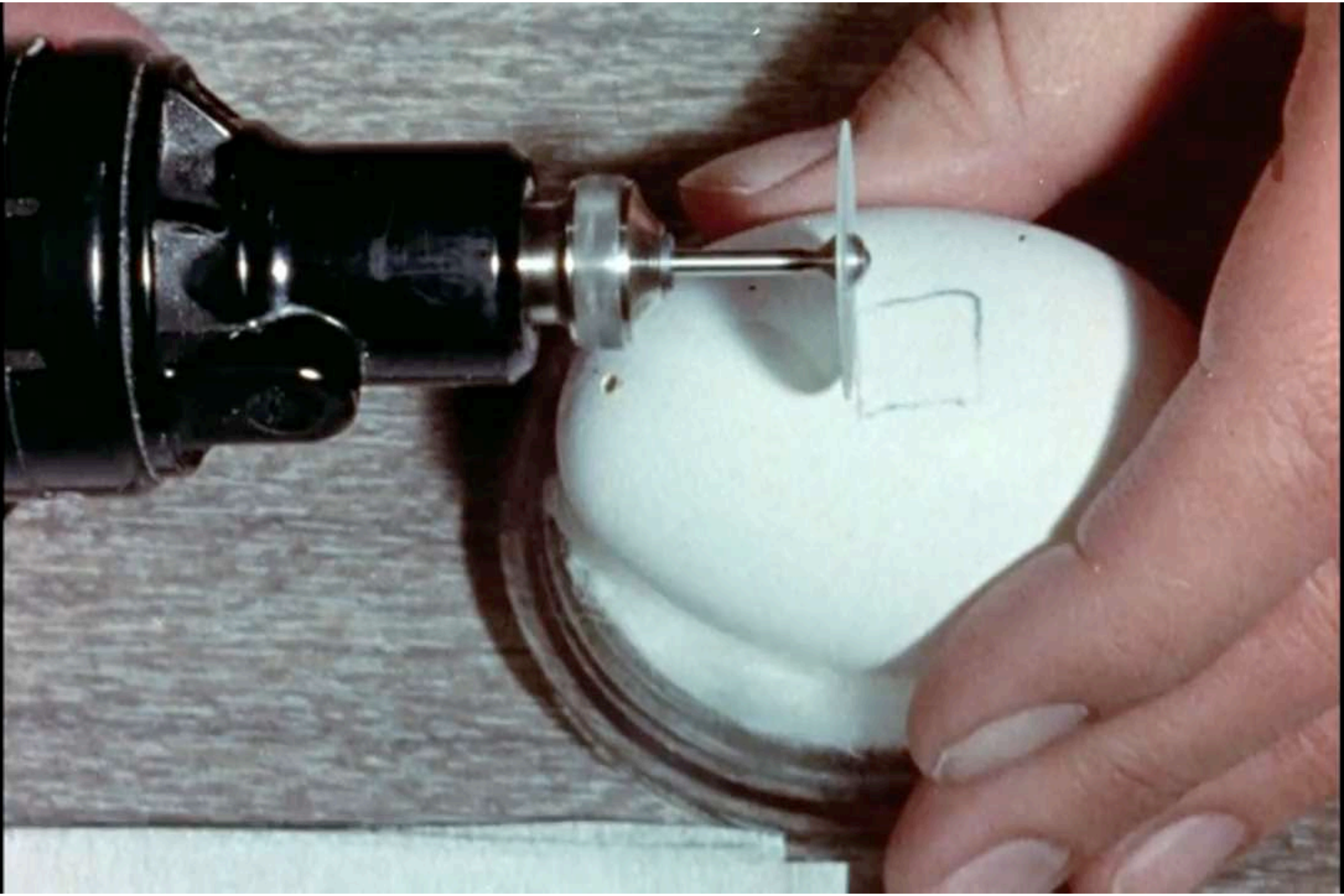
- In 1951, Theiler was awarded the Nobel Prize in Physiology or Medicine for “discoveries concerning yellow fever and how to combat it”, the first and only time the prize has been awarded for a vaccine

Viruses - vaccination

- Viruses are obligate intracellular pathogens - cannot multiply on their own, need a live host cell and its machinery to proliferate
- In the 1900s, live animals and/or animal embryos were used to culture viruses
- Later viruses were grown using animal cell cultures - still the most efficient method

Viruses - vaccination

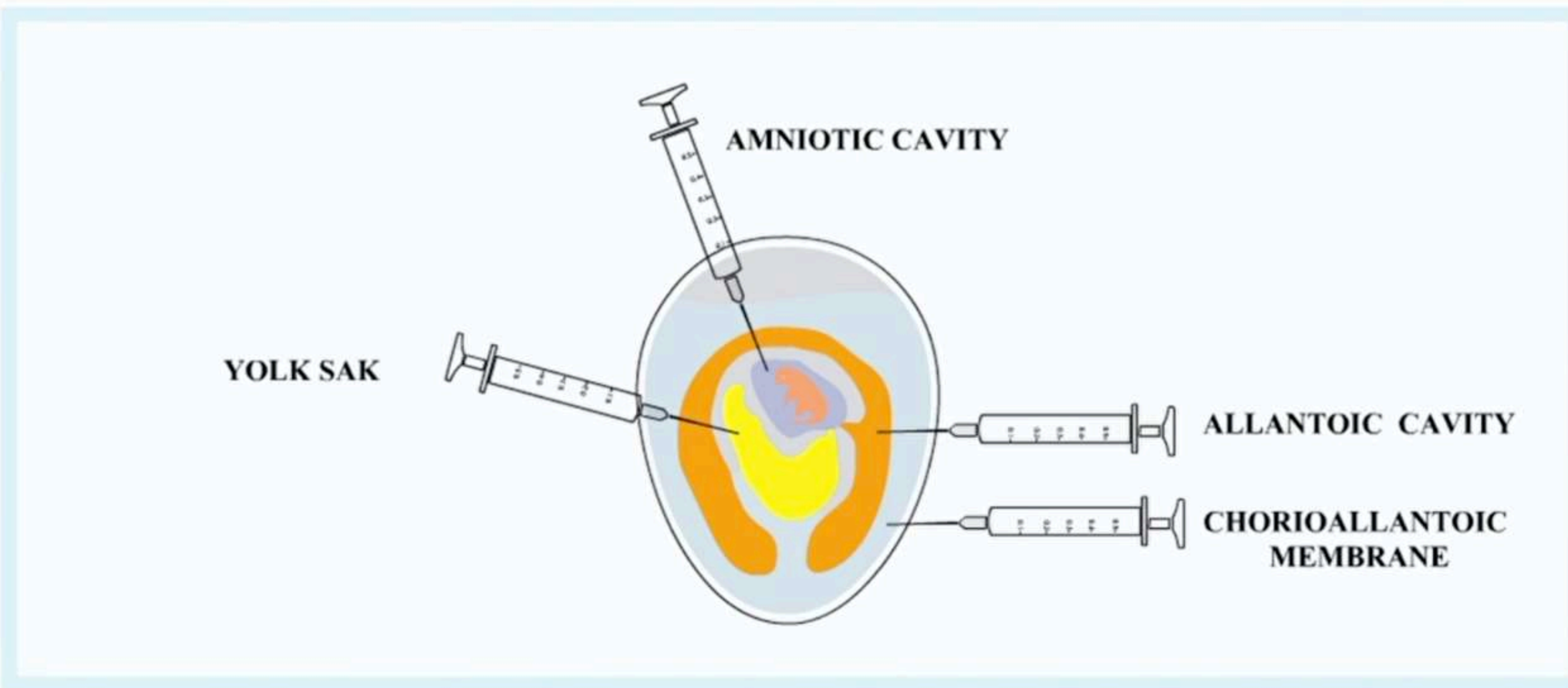
In 1930s, Ernest Goodpasture, a US pathologist and medic, grew pure viruses in culture by infecting fertilized chicken eggs



https://www.youtube.com/watch?v=NTUHX1mRfiE

Viruses - vaccination

Virus inoculation into chicken eggs - live embryo culture of viruses



https://medschool.vanderbilt.edu/md-admissions/an-egg-drill-and-its-connection-with-medical-discovery-and-advancement/

Viruses - vaccination

Poliomyelitis - caused by an enterovirus that in rare cases invades the nervous system and damages motor neurons, causing permanent disability, paralysis or death

In 1908, Karl Landsteiner and Irwin Popper showed that polio was spread by a virus:

- Injected monkeys with a suspension of spinal cord from a polio patient
- The suspension was bacteriologically sterile
- However, following inoculation the monkeys exhibited lesions in the spinal cord similar to those seen in humans with poliomyelitis
- Monkeys also developed paralysis of legs

Polio virus could not be grown in Chicken eggs:

- Albert B. Sabin and Peter K. Olitsky grew poliovirus in fragments of human embryonic brain
- Risk of using this for vaccine production was neurological damage in recipients

Viruses - vaccination

Thomas H. Weller, John F. Enders and Frederick C. Robbins
(1954 Nobel Prize in Physiology or Medicine)

Not for vaccine development, but "for their discovery of the ability of poliomyelitis viruses to grow in cultures of various types of tissue."

- grew the human Lansing strain of human poliomyelitis virus in skin, muscle and connective tissue from human fetal arms and legs
- also produced large quantities of human polio virus in cultures of fetal intestinal tissue
- succeeded in culturing human poliovirus in cells other than neurons

~1950s two polio vaccines developed:

- an injected vaccine containing inactivated virus, originally developed by Jonas Salk
- an oral vaccine containing live attenuated virus developed by Albert Sabin and colleagues

Viruses - vaccination

Whole virion, which has “forgotten” its original virulence features is used to trigger an initial immune response

This initial response is remembered by the host body as “immune memory” and stored in a dormant format

The stored, dormant “immune memory” reactivates when a new infection occurs with a virulent strain

Is the whole virion necessary to acquire immunity/ protection against viral infections?